

RECOMMENDED MATERIAL · RANK #1

Micro-Perforated Film

Breathable film · Recyclable · Not biodegradable · Relative cost: Medium

95.8

SUITABILITY SCORE / 100

Micro perforated film (1D/8) offers controlled O2/CO2 exchange and sufficient moisture for tomatoes, matching the MAP suitability rating and chilled storage compatibility. It is suitable for non hermetic packaging while maintaining food contact compliance.

Suitability score = weighted rule-based compatibility score; not a model accuracy figure.

Input Parameters

Parameter	Value	Parameter	Value
Food	Tomato	Category	Fresh Produce
Desired shelf life	10 days (target only)	Storage	chilled at 8 °C
Transportation	Refrigerated	Priority	Shelf Life
Recommendation mode	Groq: openai/gpt-oss-120b	Confidence	Medium

Food Intelligence - Typical / Estimated Profile

- Groq reasoning grounded in the food and packaging knowledge bases.
- Qualitative evidence confidence, not an accuracy score; no laboratory or shelf-life validation. Sustainability is an indicative end-of-life proxy, not an LCA.

Property	Typical value	Provenance
Ph	4-4.9	Prototype reference range/profile
Fat	0.1-0.5%	Prototype reference range/profile
Acidity	Acidic to mildly acidic; verify batch pH	Prototype reference range/profile
Moisture	90-95%	Prototype reference range/profile
Respiration	High (temperature/maturity dependent)	Prototype reference range/profile
Co2 Sensitivity	Relevant: commodity-specific tolerance must be verified	Prototype reference range/profile
Oxygen Sensitivity	Medium	Prototype reference range/profile
Moisture Sensitivity	High	Prototype reference range/profile

Storage Warnings and Data Assumptions

- Maturity-dependent: mature-green fruit generally needs 12.5-15 °C; firm-ripe fruit may tolerate 7-10 °C briefly. An 8 °C / 10-day target needs chilling-injury validation.
- Desired shelf life is a target, not a validated prediction or safety guarantee.
- Requested 8 °C is outside the typical 10-15 °C storage band. Packaging cannot compensate for inappropriate storage.
- Food-property bands are typical/reference or AI-estimated values, not measurements of your food.

- Reference-range midpoints are used internally for weighted scoring; they are not laboratory measurements.
- Relative humidity is a scenario assumption (85% RH), inferred from the selected conditions, not measured.

Packaging Specifications

Property	Value	Target / note
OTR	10,000 – 2,00,000 cm ³ /m ² ·day	Very high / breathable (> 5,000 cm ³ /m ² ·day or micro-perforated) - Meets target
WVTR	10 – 60 g/m ² ·day	Low-moderate (5-60 g/m ² ·day) to limit weight loss without condensation - Meets target
Recommended thickness	30 µm	Material range 20 – 50 µm
Sealability	Good (4/5)	
Gas permeability	Very high (breathable) (5/5)	
Mechanical strength	Moderate (3/5)	
MAP suitability	Excellent (5/5)	
Suggested format	Micro-perforated LDPE film	

Indicative reference ranges, not lab measurements: nominal OTR 23 °C / 0% RH; WVTR 38 °C / 90% RH. Validate the complete pack, lid and perforations.

Why This Material?

- + Breathable micro perforations enable proper gas exchange for high respiration
- + Moisture transmission matches high moisture sensitivity of tomatoes
- + Suitable for chilled storage and fresh produce category
- + High MAP suitability rating

Score Breakdown

Criterion	Criterion score	Effective weight	Contribution
Food compatibility	100%	20.8%	20.8
Storage compatibility	100%	16.7%	16.7
Barrier requirements	93%	25.0%	23.1
Shelf-life suitability	100%	25.0%	25.0
Cost	85%	4.2%	3.5
Sustainability	75%	4.2%	3.1
MAP suitability	85%	4.2%	3.5

Key Decision Factors

Factor	Input	Derived requirement	Influence
Respiration rate	High (inferred profile)	Controlled gas exchange required (target permeability ~5/5)	High
Moisture sensitivity	High (inferred profile)	Moisture barrier >= 4.0/5 (high sensitivity)	High
Oxygen sensitivity	Medium (inferred profile)	Oxygen barrier ~2.5/5 (medium sensitivity, capped because	Medium

Factor	Input	Derived requirement	Influence
		respiring produce needs O2 ingress)	
Storage temperature	8 °C (chilled)	Material must be rated for chilled conditions	Medium
Relative humidity	85% (scenario assumption)	High humidity around respiring produce: breathability reduces condensation risk	High
Shelf-life target	10 days (short)	Protection level ~2.5/5 required	Medium
Transportation	Refrigerated	Mechanical strength >= 3/5 suggested	Low
MAP requirement	Not sure	Passive MAP (respiration-driven equilibrium atmosphere) - suggested by engine	Medium

Alternative Materials

Rank	Material	Score	Main advantage	Main trade-off
#2	LDPE	91.6	LDPE provides a flexible, sealable film with adequate gas permeability for tomatoes, though less precise control than micro perforated film	Gas permeability (4/5) is below the ~5/5 needed: risk of CO2 build-up
#3	PP	88.7	PP offers better moisture barrier but lower gas permeability, making it a secondary option if higher strength is needed	Gas permeability (3/5) is below the ~5/5 needed: risk of CO2 build-up

Trade-offs to Consider

- Micro-Perforated Film: Moisture barrier (3/5) is below the required ~4.0/5
- LDPE scores higher on cost but ranks 4.2 points lower overall
- PP scores higher on cost but ranks 7.1 points lower overall

Full Ranking and Filtered-out Materials

Rank	Material	Suitability score
#1	Micro-Perforated Film	95.8
#2	LDPE	91.6
#3	PP	88.7
#4	PET	74.0

Excluded material	Reason(s)
HDPE	No grounded food-category compatibility for this structure
Metallized Film	Near-hermetic barrier: with high respiration this risks anaerobic conditions and off-flavours; Respiring produce requires ventilation; an unmodified hermetic barrier is unsuitable; No grounded food-category compatibility for this structure
Aluminum Foil Laminate	Near-hermetic barrier: with high respiration this risks anaerobic conditions and off-flavours; Respiring produce requires ventilation; an unmodified hermetic barrier is unsuitable; No grounded food-category compatibility for this structure
Biodegradable Film	Moisture barrier is insufficient; sustainability cannot override technical suitability
High-Barrier Laminate	Near-hermetic barrier: with high respiration this risks anaerobic conditions and off-flavours; Respiring produce requires ventilation; an unmodified hermetic barrier is unsuitable; No grounded food-category compatibility for this structure

Excluded material	Reason(s)
Paper-Based Ventilated Bag	Moisture barrier is insufficient; sustainability cannot override technical suitability
PP Cup with Sealed Lid	No grounded food-category compatibility for this structure
Opaque HDPE Bottle	Packaging format is not suitable for containing this food form; No grounded food-category compatibility for this structure

Disclaimer

Values shown are reference/estimated ranges and should be validated against laboratory data and packaging standards before commercial use. This prototype does not predict or guarantee shelf life. No chemical values are measured from images.